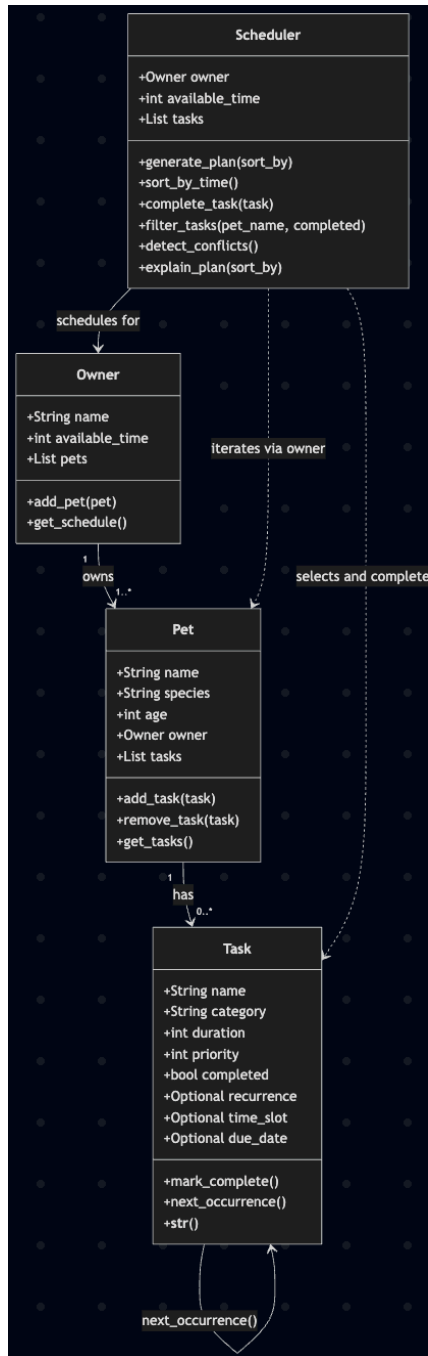




```

classDiagram
    class Owner {
        +String name
        +int available_time
        +List pets
        +add_pet(pet)
        +get_schedule()
    }
  
```

```

class Pet {
    +String name
    +String species
    +int age
    +Owner owner
    +List tasks
    +add_task(task)
    +remove_task(task)
    +get_tasks()
}

class Task {
    +String name
    +String category
    +int duration
    +int priority
    +bool completed
    +Optional recurrence
    +Optional time_slot
    +Optional due_date
    +mark_complete()
    +next_occurrence()
    +__str__()
}

class Scheduler {
    +Owner owner
    +int available_time
    +List tasks
    +generate_plan(sort_by)
    +sort_by_time()
    +complete_task(task)
    +filter_tasks(pet_name, completed)
    +detect_conflicts()
    +explain_plan(sort_by)
}

Owner "1" --> "1..*" Pet : owns
Pet "1" --> "0..*" Task : has
Task --> Task : next_occurrence()
Scheduler --> Owner : schedules for
Scheduler ..> Pet : iterates via owner

```

```
Scheduler ..> Task : selects and completes
```